



PROJECTS & TECHNICAL CENTERS (P&TC) PROJECTS & CONSTRUCTION (BAB/NEB)

EPC FOR NEB MISCELLANEOUS MODIFICATIONS AND ENHANCEMENTS PROJECT

ADNOC Onshore Contract No.: 4700018825

ADNOC Onshore Project No.: P30319

RSME	
EC/EM	PM
<i>[Signature]</i>	<i>[Signature]</i>

(PKG-34)

MECHANICAL DATA SHEET FOR CORROSION INHIBITOR PACKAGE (CLS-28 BB & CLS-30 DD) AT SHANAYEL SN-45 & SN-40

(TAG NO. 65-72-U-045P-01 & 65-21-U-6230-01)

This Document is developed from the FEED Doc. No: P30316-30-99-27-0604, rev. 2					
		<i>Shinobu N.R.</i>	KR.	<i>Ujjwal</i>	
B	05-Jan-2024	SNR	KR	UM	Issued for Review
A	25-Oct-2023	SNR	KR	UM	Issued for Review
REV.	DATE	ORIGINATOR	REVIEWED	APPROVED	DESCRIPTION
THIS DOCUMENT IS INTENDED FOR USE BY ADNOC AND ITS NOMINATED CONSULTANTS, CONTRACTORS, MANUFACTURERS AND SUPPLIERS.					
ORIGINATOR: China Petroleum Engineering Co., Ltd, Dubai			CONTRACTOR: ROBTSTONE (ME) LLC		

ADNOC Onshore Document No. :

P30319-65-21-34-1601

Revision :

B

ORIGINATOR Project No.:

P123091-MES-0016

Date:

05-Jan-2024

ADNOC Onshore Project No.:

P30319

Page:

1 of 10

Security Code: 5 – Public

Abu Dhabi Company for Onshore Petroleum Operations Ltd. شركة أبوظبي للعمليات البترولية البرية المحدودة.
PO Box 270, Abu Dhabi, UAE أبوظبي، الإمارات العربية المتحدة ص ب 270
adnoc.ae

أدنوك البرية
ADNOC Onshore

**MECHANICAL DATA SHEET FOR CORROSION INHIBITOR PACKAGE
(CLS-28 BB & CLS-30 DD) AT SHANAYEL SN-45 & SN-40**

Document No. P30319-65-21-34-1601

Rev B

Project No. P30319

Page 2 of 10



The table below is a brief summary of the most recent revisions to this document. Details of all revisions are held on document by the issuing department.

Sr. No.	Rev. No.	Issue No.	Date of issue	Description of revision
1	A	1	25-Oct-2023	ISSUED FOR REVIEW
2	B	2	05-Jan-2024	ISSUED FOR REVIEW

MECHANICAL DATA SHEET FOR CORROSION INHIBITOR PACKAGE(CLS-28 BB & CLS-30 DD) AT SHANAYEL SN-45 & SN-40

Project No. P30319

REV.: B

PAGE: 3 OF 10

Document No.: P30319-65-21-34-1601

Date: 05-Jan-2024



STORAGE TANK

1	APPLICABLE TO: ☐ STUDY ☒ INQUIRY ☐ PURCHASE ☐ AS-BUILT ☐ APPROVAL				REVISION
2	Client: ADNOC ONSHORE		Package Tag No.: 65-72-U-045P-01 / 65-21-U-6230-01		
3	Facility: SHANAYEL		Service: Corrosion Inhibitor Injection		
4	Location: CLS 28(BB) / CLS 30(DD)		No. Req'd: 2		
5	Manufacturer: As per approved Vendor list		Model & Type: By Vendor		
6					
7					
8	GENERAL				
9	APPLICABLE SPECIFICATIONS: AGES-SP-05-011				
10	CODES AND STANDARDS: ASME Sec VIII Div 1				
11	CAPACITY: NOM. 0.29 m ³ , NET WORKING 0.13 m ³				
12	SHELL ID x TL: Refer Table-B TANK PROTECTION: IP 53				
13	PROCESS ENVIRONMENT: CORROSION INHIBITOR				
14	PROCESS CONTENTS: Corrosion Inhibitor				
15	VAPOR PRESSURE: 0.2388 hPa @ 20 °C				
16	DESIGN AND OPERATING CONDITIONS				
17					
18	TEMPERATURE, MAXIMUM 85 °C 58 °C				
19	TEMPERATURE, MINIMUM 4 °C 4 °C				
20	INTERNAL PRESSURE Full of liquid + 0.5 barg ATM barg				
21	EXTERNAL PRESSURE 0 barg 0 barg				
22	SPECIFIC GRAVITY OF PROCESS FLUID 1.013 AT 20 °C				
23	GASES IN VAPOR SPACE: Air				
24					
25	CORROSION ALLOWANCE: SHELL 0 mm; BOTTOM 0 mm				
26	ROOF 0 mm; STRUCTURAL 0 mm				
27	NOZZLE 0 mm; INTERNALS 0 mm				
28	PUMP RATE: IN By vendor m ³ /HR; OUT By vendor m ³ /HR				
29	WIND: ASCE 7-16 (Note-8.17) IMPORTANCE FACTOR 1.0				
30	EXPOSURE C BASIC WIND SPEED 56.3 m/s				
31	SEISMIC: Note-8.17 IMP. FACT. Note-8.17 SOIL PROFILE Note-8.17				
32	SNOW LOAD NA Mpa ROOF LIVE LOAD Mpa				
33	FOUNDATION TYPE ☐ CONCRETE ANCHORED ☐ EARTH				
34	☐ CONCRETE RINGWALL ☒ OTH MOUNTED ON SKID				
35					
36	MAXIMUM BASE MOMENT By Vendor N-m, DUE TO Wind				
37	MAXIMUM BASE SHEAR By Vendor N, DUE TO Wind				
38	MAXIMUM BASE MOMENT By Vendor N-m, DUE TO Seismic				
39	MAXIMUM BASE SHEAR By Vendor N, DUE TO Seismic				
40					
41	MATERIALS				
42	PRODUCT FORM PRESSURE PARTS NON-PRESSURE PARTS				
43	EXTERNAL INTERNAL				
44	PLATE / PAD SA240-316L SA240-316L SA240-316L				
45	FORGING SA182-F316L SA182-F316L SA182-F316L				
46	NOZZLE NECK SA312-TP316L --- SA312-TP316L				
47	STUD SA-193 B8M SA-193 B8M SA-193 B8M				
48	HEX. NUT SA-194 Gr. 8M SA-194 Gr. 8M SA-194 Gr. 8M				
49	ROD AND BAR --- ---				
50	PIPE FITTINGS: SA403-WP316L-S SA403-WP316L-S SA403-WP316L-S				
51	GASKET 316L Spiral Wound w/graphite seal				
52	SUPPORTS / BASE PLATE SA240-316L S 235 JR / S 275 JR				
53					
54					
55					
56	Notes: 3.1 Nozzles to be 2" minimum. Handhole to be 8" minimum.				
57	3.2 All testing & NDE shall be as per Code, Specifications and approved ITP.				
58	3.3 Level instrument to cover the entire height of the vessel.				
59	3.4 Vent to be provided with flame arrestor				
60	3.5 Nozzle projection as per SHELL Std Drg. no: S 10.101				
61	3.6 Refer to P&ID No: P30319-65-72-08-1602 & P30319-02-21-08-1695.				
62	3.7 Vessel shape indicated in sketch (sht-4) is indicative with table of nozzle and shall be confirmed by Vendor.				
63	3.8 All Flanges shall be as per ASME 16.5. Flange bolt holes shall straddle centerline.				
64	3.9 All nozzles of 2" NPS size shall be provided with 2 stiffeners - 30 x 6T at 90° apart.				
65	3.10 Tank working volume is calculated based on 28 days of continuous consumption.				
66	3.11 Vendor to select suitable design pressure based on flame arrestor pressure drop.				
67	3.12 Positive Material Identification (PMI) & Ferrite testing shall comply with requirement in Project Specification AGES-GL-13-002.				
68	3.13 Stainless steels SS316, SS316L shall pass the Inter Granular Corrosion (IGC) test conducted by manufacturer in accordance with				
69	ASTM A262 Practice E.				
70	3.14 Austenitic SS (316L type) shall meet the chemical requirements of X3CrNiMo17-3-3 as per BS10088, with a minimum PREN number of 27.5.				
71	3.15 Vessel shall be supplied with suitable preservation desiccant as per Preservation & Export Packing Specification (AGES-SP-07-011).				

MECHANICAL DATA SHEET FOR CORROSION INHIBITOR PACKAGE(CLS-28 BB & CLS-30 DD) AT SHANAYEL SN-45 & SN-40

Project No. P30319

REV.: B

PAGE: 4 OF 10

Document No.: P30319-65-21-34-1601

Date: 05-Jan-2024



STORAGE TANK

1	APPLICABLE TO: ☐ STUDY ☒ INQUIRY ☐ PURCHASE ☐ AS-BUILT ☐ APPROVAL				REVISION
2	Client: ADNOC ONSHORE	Package Tag No.: 65-72-U-045P-01 / 65-21-U-6230-01			
3	Facility: SHANAYEL	Service: Corrosion Inhibitor Injection			
4	Location: CLS 28(BB) / CLS 30(DD)	No. Req'd: 2			
5	Manufacturer: As per approved Vendor list	Model & Type: By Vendor			
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					
18					
19					
20					
21					
22					
23					
24					
25					
26					
27					
28					
29					
30					
31					
32					
33					
34					
35					
36					
37					
38					
39					
40					
41					
42					
43					
44					
45					
46					
47					
48					
49					
50					
51					
52					
53					
54					
55					
56					
57					
58					
59					
60					
61					
62					
63					
64					
65					
66					
67					

(*) Vendor to confirm

- Vendor to provide saddle support with sliding plate.
- Four lifting lugs shall be provided for horizontal tank and shall be performed local stress analysis.

NOZZLE SCHEDULE								
MARK	NO. REQ'D	SIZE	RATING	TYPE	FACE	SERVICE	LOCATION	REMARKS
M1	1	8"	150#	WN	RF	Handhole	On Shell	Note 4.1
N1	1	2"	150#	WN	RF	Liquid Inlet	Top cover	Note 4.3
N2	1	2"	150#	WN	RF	Vent	Top cover	Note 3.4
N3	1	2"	150#	WN	RF	Level Transmitter	Top cover	
N4A/B	2	2"	150#	WN	RF	Level Gauge	Shell	Note 3.3
N5	1	2"	150#	WN	RF	Liquid Out	Bottom on Shell Head	Note 4.4 , 4.2
N6A/B	2	2"	150#	WN	RF	Level Transmitter	Shell	Note 3.3
N7	1	2"	150#	WN	RF	Return Line	Top cover	Note 4.3
N8	1	2"	150#	WN	RF	Drain	Bottom on Shell Head	Note 4.4, 4.2
N9	1	2"	150#	WN	RF	Calibration pot	Top cover	

Notes:

- Handhole shall be provided with blind, gasket and fasteners as per Shell std. S 22.010 / S 10.070.
- Level nozzles shall be jig set.**
- Nozzle shall be provided with internal dip pipes with support at bottom to avoid vibration during filling / transportation with anti-siphon hole in internal pipe of N1 & N7.
- Vortex break as per Shell std. S 10.010
- Local load for nozzles shall be considered as per requirement in AGES-SP-06-001 Design Criteria for Static Equipment, Appendix 2.**

MECHANICAL DATA SHEET FOR CORROSION INHIBITOR PACKAGE(CLS-28 BB & CLS-30 DD) AT SHANAYEL SN-45 & SN-40

Project No. P30319

REV.: B

PAGE: 5 OF 10

Document No.: P30319-65-21-34-1601

Date: 05-Jan-2024



CONTROLLED VOLUME PUMP (API 675)

SI UNITS (bar)

1	APPLICABLE TO: ☒ PROPOSAL ☐ PURCHASE ☐ AS BUILT				REVISION
2	FOR ADNOC ONSHORE		PACKAGE TAG No.	65-72-U-045P-01 / 65-21-U-6230-01	
3	LOCATION / FACILITY	CLS 28(BB) / CLS 30(DD) SHANAYEL	NO. OF PUMPS REQUIRED	TWO (1W + 1S) PER SKID	
4	SERVICE	Corrosion Inhibitor Injection	MODEL	By Vendor	
5	MANUFACTURER	As per approved Vendor list		SERIAL NO.	
6	By Vendor				
7	NOTE: ☐ INDICATES INFORMATION TO BE COMPLETED BY PURCHASER ☐ BY MANUFACTURER				
8	GENERAL				
9	NO. MOTORS DRIVEN	One per each pump	OTHER DRIVER TYPE		
10	PUMP ITEM NO'S	65-72-P-045P-01	PUMP ITEM NO'S	65-21-P-6230-01	
11	MOTOR ITEM NO'S	65-72-PM-045P-01	DRIVER ITEM NO'S	65-21-PM-6230-01	GEAR ITEM NO'S
12	MOTOR PROVIDED BY	Package Vendor	DRIVER PROVIDED BY		GEAR PROVIDED BY
13	MOTOR MOUNTED BY	Package Vendor	DRIVER MOUNTED BY		GEAR MOUNTED BY
14	MOTOR DATA SHEET NO.	VTA	DRIVER DATA SHEET NO.		GEAR DATA SHEET NO.
15	*Vendor Must Complete				
16	☐ OPERATING CONDITIONS (EACH PUMP)				
17	☒ LIQUID				
18	● CAPACITY @ PT (L/h):				
19	RATED	Refer Table-B	MINIMUM	0	
20	● DISCHARGE PRESSURE (barg) :	Refer Table-B			
21	● MAXIMUM ALLOW. PRESSURE:	Note 5.1			
22	○ SUCTION PRESSURE (barg) :	VTA			
23	MAXIMUM	ATM	MINIMUM	VTA	
24	● DIFFERENTIAL PRESSURE (barg) :	Refer Table-B			
25	RATED	MINIMUM			
26	● NPSH AVAILABLE (m)	Per Vendor Design			
27	WITHOUT ACCELERATION HEAD	ACTUAL			
28	● TURNDOWN RATIO	0-100%			
29	☒ PERFORMANCE (VTA)				
30	NUMBER OF FEEDS	Note-5.2	CAPACITY Normal / Max. (m³/h)	Table-B	
31	NPSH REQUIRED (m)		RATED CAPACITY (m³/h)		
32	BkW RATED		AT RELIEF SETTING		
33	PLUNGER SPEED (strokes/min)		DESIGN MAX		
34	PLUNGER DIAMETER (mm)		LENGTH OF STROKE (mm)		
35	PUMP HEAD (bar)				
36	MAXIMUM PRESSURE (barg)				
37	HYDRO TEST PRESSURE (barg)		1.5 X MAWP		
38	MAX DISCH. PRESS. W/ JOB DRIVER (barg)				
39	MAX BkW BASIS GEAR STRENGTH				
40	☒ CONSTRUCTION				
41	CONNECTIONS	SIZE	ANSI RATING	FACING	POSITION
42	SUCTION	BY VENDOR	2500 #	RF	
43	DISCHARGE	BY VENDOR	2500 #	RF	
44	FLUSH				
45	JACKET				
46	LIQUID END	☐ JACKET REQ'D			
47	TYPE	☒ DOUBLE DIAPHRAGM ☐ PLUNGER			
48	DIAPHRAGM DIA (mm)	By Vendor NO REQ. Two			
49					
50	VALVES/FEED	SUCTION	DISCHARGE		
51	TYPE	Ball	Ball		
52	NUMBER	Double	Double		
53					
54	Notes:				
55	5.1) Max allowable pressure shall be selected suitable to piping class indicated in the respective P&IDs (Doc. No. P30319-65-72-08-1702).				
56	5.2) Refer to P&ID No: P30319-65-72-08-1602 & P30319-02-21-08-1695.				
57	5.3) Material of construction for double diaphragm type metering pump shall be suitable for chemical used.				
58	5.4) Vendor to provide the pump power required.				
59					
60					
61					
62					
63					
64					
65					
66					
67					

MECHANICAL DATA SHEET FOR CORROSION INHIBITOR PACKAGE (CLS-28 BB & CLS-30 DD) AT SHANAYEL SN-45 & SN-40

Project No. P30319

REV.: B

PAGE: 6 OF 10

Document No.: P30319-65-21-34-1601

Date: 05-Jan-2024



CONTROLLED VOLUME PUMP (API 675)

SI UNITS (bar)

1	APPLICABLE TO: ☒ PROPOSAL ☐ PURCHASE ☐ AS BUILT		REVISION	
2	FOR ADNOC ONSHORE	PACKAGE TAG No. 65-72-U-045P-01 / 65-21-U-6230-01		
3	LOCATION / FACILITY CLS 28(BB) / CLS 30(DD) SHANAYEL	NO. OF PUMPS REQUIRED TWO (1W + 1S) PER SKID		
4	SERVICE Corrosion Inhibitor Injection MODEL By Vendor	SIZE AND TYPE Double Diaphragm Metering Pump		
5	MANUFACTURER As per approved Vendor list	SERIAL NO. By Vendor		
6				
7	MATERIALS			
8	LIQUID END SS 316 L	CONTROLS		
9	CONTOUR PLATE SS 316 L			
10	HYDRAULIC DIAPHRAGM PTFE			
11	PROCESS DIAPHRAGM PTFE			
12	PLUNGER			
13	LANTERN RING NA PACKING GLAND NA	CAPACITY CONTROL: ☒ MANUAL ☐ REMOTE ☒ LOCAL CONTROL RANGE: 0-100% ☐ AUTOMATIC MECHANISM: ☐ SPEED CONTROL ☒ STROKE CONTRL SIGNAL: ☐ PNEUMATIC ☐ ELECTRONIC ☐ REMOTE ☒ LOCAL, MICROMETRIC CALIBRATION CAPACITY GRADUATIONS 0 - 100 % (0-20 MM)		
14	PACKING NA VALVE SS 316 L			
15	VALVE BODY SS 316 L VALVE SEAT SS 316 L			
16	VALVE GASKET SS 316 L			
17	RELIEF VALVE SS 316 L BACK PRESSURE VALVE SS 316 L			
18	COUPLING GUARD Non-Sparking	OTHER PURCHASE REQUIREMENTS NAMEPLATE UNITS ☐ U.S. CUSTOMARY ☒ SI ☒ VENDOR FURNISHED PROCESS PIPING All on skid mounted piping / valves by package vendor ☐ VENDOR REVIEW PIPING DRAWINGS ☒ VENDOR FURNISHED PULSATION SUPPRESSION DEVICES ☒ VENDOR FURNISHED RELIEF VALVE ☐ INTERNAL ☒ EXTERNAL ☒ RELIEF VALVE SETTING (barg) 2 nos (Int / Ext) as per API 675 ☒ VENDOR FURNISHED BACK-PRESSURE VALVE ☒ DOUBLE CHECK VALVES REQUIRED ☐ OIL-FILLED PRESSURE GAUGES REQUIRED ☒ VENDOR FURNISHED CONTROL PANEL ☒ BASE PLATE PREPARED FOR EPOXY GROUT ☒ PROVIDE TECHNICAL DATA MANUAL ☐ ☐		
19	FRAME Cast Iron BASE FRAME CS Galvanized			
20	FASTENERS (PIPING) SA 193 GR B8M / SA 194 GR 8M (Note 8.7)			
21				
22	SPECIAL MATERIAL TESTS (2.13.1.3)			
23	☐ LOW AMBIENT TEMPERATURE MATERIALS TESTS (2.13.5)			
24				
25				
26	QA INSPECTION AND TEST (Note 8.1)			
27	☒ COMPLIANCE WITH INSPECTORS CHECK LIST	PREPARATION FOR SHIPMENT ☐ DOMESTIC ☒ EXPORT ☒ EXPORT BOXING ☒ OUTDOOR STORAGE MORE THAN 12 MONTHS WEIGHTS (kg) ☒ PUMP ☐ BASE ☐ GEAR ☐ DRIVER DRIVERS ☒ MOTOR: See Motor Data Sheet ☐ MANUFACTURER ☐ TYPE ☐ FRAME NO. ☐ RATED POWER ☐ POWER AT RV SETTING: ☒ CONSTANT SPEED ☐ VARIABLE SPEED ☐ TRANSMISSION: ☒ Direct ☐ Gear ☐ Other ☒ kW 0.5 (VTC) RPM ☒ VOLTS * (VAC) PHASE * ☒ HERTZ * SERVICE FACTOR ☒ ENCLOSURE Exde, IP55 ☐ OTHER (SEE SEPARATE DATA SHEETS) ☐ GAS DRIVEN ☐ STEAM TURBINE ROTATION FROM DRIVE END ☐ CW ☐ CCW (*) 415 V AC 50 Hz 3 Phase motor		
28	☒ CERTIFICATION OF MATERIALS			
29	☒ FINAL ASSEMBLY CLEARANCES			
30	☐ SURFACE AND SUBSURFACE EXAMINATIONS			
31	☒ RADIOGRAPHY			
32	☒ ULTRASONIC			
33	☒ MAGNETIC PARTICLE			
34	☒ LIQUID PENETRANT			
35	☒ Ferrite Testing			
36	☒ Positive Material Identification (PMI)			
37	☒ CLEANLINESS PRIOR TO FINAL ASSEMBLY			
38	☒ HARDNESS OF PARTS, WELDS & HEAT AFFECTED ZONES			
39	☐ FURNISH PROCEDURES FOR OPTIONAL TESTS			
40	TESTS	REQ'D	WIT	OBS
41	Dimensional Check	☒	☐	☐
42	Hydrostatic	☒	☒	☐
43	Steady State Accuracy	☒	☒	☐
44	Repeatability	☒	☒	☐
45	Linearity	☒	☐	☐
46	Mechanical Run Test	☒	☐	☒
47	Functional Tests	☒	☒	☐
48	Diaphragm rupture detection simulation test	☒	☐	☒
49				
50	LUBRICATION FLUID			
51	☐ CRANKCASE ☐ INTERMEDIATE			
52	☐ HYDRAULIC FLUID			
53	ACCESSORIES			
54	☐ SPEED REDUCER MANUFACTURER			
55	☐ INTEGRAL ☐ SEPARATE			
56	MODEL			
57	RATIO			
58	☐ BASE PLATE UNDER			
59	☒ COUPLING: MAKE MODEL			
60	☒ Flexible ☐ Rigid ☐ Gear ☐ Disc ☐ Spacer			
61				
62				
63				
64				
65				
66				
67				

MECHANICAL DATA SHEET FOR CORROSION INHIBITOR PACKAGE(CLS-28 BB & CLS-30 DD) AT SHANAYEL SN-45 & SN-40

Project No. P30319

REV.: B

PAGE: 7 OF 10

Document No.: P30319-65-21-34-1601

Date: 05-Jan-2024



APPURTENANCES

1	APPLICABLE TO:	☒ PROPOSAL	☐ PURCHASE	☐ AS BUILT	PACKAGE TAG NO.	65-72-U-045P-01 / 65-21-U-6230-01	REVISION	
2	CLIENT	ADNOC ONSHORE			SERVICE	Corrosion Inhibitor Injection		
3	LOCATION / FACILITY	CLS 28(BB) / CLS 30(DD) SHANAYEL						
4	PACKAGE MANUFACTURER	As per approved Vendor list			MODEL	By Vendor		
5	VENDOR SHALL COMPLETE EACH SECTION BELOW FOR EACH ITEM (ADD SHEET IF NEEDED) (*)							
6	PULSATION DAMPENERS			CALIBRATION POT				
7				SERVICE TAG NO(S)				
8	TAG NO(S)				BODY MATERIALS			
9	MFR.	MODEL			GRADUATED CALIBRATION STRIP			
10	SIZE				STAINLESS STEEL GUARD			
11	MATERIALS OF CONSTRUCTION	316SS/L			CONNECTIONS			
12	GAS CHARGED WITH FLEXIBLE MEMBRANE	REQUIRED			VENT			
13	CONNECTIONS				NOTE: SUPPORTED INDEPENDENTLY FROM PIPE WORK			
14	SS PIPING, TUBING & VALVES			COUPLINGS & COUPLING GUARD				
15	PIPING & FITTING MATERIALS	SS 316L			SIZE (MM) VTA			B
16	FLANGE FACE SURFACE FINISH	REFERENCE PIPING SPECIFICATION			TYPE Non-Sparking			B
17	TUNING & FITTINGS	SS 316L			MFR. VTA			B
18	NOZZLE LOADS				MATERIAL VTA			B
19	VALVE TYPES/CONNECTION DETAIL	As per piping material class & Tubing Valves						
20	VALVE BODY MATERIALS	SS316L			SEAL MATERIAL VTA			B
21	VALVE TRIM MATERIALS	SS316L						
22								
23	LOCAL CONTROL PANEL / JB / INSTRUMENTATION			IN-LINE STRAINER (VTA)			B	
24	As per Typical P&ID (Vendor to provide detailed P&ID along with offer)			SERVICE				
25	MFR.	MODEL			MFR.			
26	TYPE				TYPE			
27	MATERIALS	SS 316L			BODY MATERIALS			
28				FILTER MESH MATERIALS				
29								
30	RELIEF VALVE, BACK PRESSURE VALVE, CHECK VALVE			SCREEN MESH SIZE				
31	As per Typical P&ID (Vendor to provide detailed P&ID along with offer)			MAXIMUM PRESSURE DROP				
32	MFR.	MODEL			3/4" VALVED & PLUGGED			
33	TYPE				BLOWDOWN CONNECTION			
34	MATERIALS	SS 316L			CONNECTIONS			
35								
36	FLAME ARRESTOR							
37	MFR.	MODEL						
38	TYPE							
39	MATERIALS	SS 316L						
40								
41	HOSE WITH ADAPTOR CONNECTION			HAND OPERATED DRUM UNLOADING PUMP				
42	MANUFACTURER/SUPPLIER				SERVICE			
43	HOSE LENGTH (M)				MFR.			
44	HOSE MATERIAL	SS 316L			MODEL			
45				MATERIAL				
46				OPERATION ELECTRIC				
47				CAPACITY (NORMAL) L/H 10000 / 2.2 kW				
48				Tag no: 44-21-P6209-07/08				
49	COMMON BASE FRAME, STRUCTURAL SUPPORT, PLATFORM LADDER, DRIP PAN - REQUIRED							
50	7.1 Positive Material Identification (PMI) & Ferrite testing shall comply with requirement in Project Specification AGES-GL-13-002.							
51								
52								
53								
54								
55								
56								
57								
58								
59								
60								
61								
62								
63								
64								
65								
66								
67								

MECHANICAL DATA SHEET FOR CORROSION INHIBITOR PACKAGE(CLS-28 BB & CLS-30 DD) AT SHANAYEL SN-45 & SN-40

Project No. P30319

REV.: B

PAGE: 8 OF 10

Document No.: P30319-65-21-34-1601

Date: 05-Jan-2024



NOTES

1	APPLICABLE TO:	☒ PROPOSAL	☐ PURCHASE	☐ AS BUILT	PACKAGE TAG NO.	65-72-U-045P-01 / 65-21-U-6230-01		REVISION		
2	CLIENT	ADNOC ONSHORE			SERVICE	Corrosion Inhibitor Injection				
3	LOCATION / FACILITY	CLS 28(BB) / CLS 30(DD)			SHANAYEL					
4	PACKAGE MANUFACTURER	As per approved Vendor list			MODEL	By Vendor	SERIAL NO. By Vendor			
5										
6	8.1. This data sheet shall be read along with Chemical Injection Package Specification AGES-SP-05-011 and AGES-SP-05-006.									
7	8.2. Separate data sheet for motor shall be submitted by Vendor during detail engineering.									
8	8.3. Pulsation dampeners at discharge of the pump shall be provided. Requirement of pulsation damper at pump suction side shall be									
9	determined by vendor.									
10	8.4. All piping material shall conform to project Piping Material Specification (Doc No. P30319-30-99-12-1602). SS316 shall be supplied in									
11	Dual Certification.									
12	8.5. Calibration pot shall be protected by steel cage.									
13	8.6. Pump flow rate shall be manually adjusted from 0 to 100% by micrometer control, while running or stationary. Loss of mechanism is									
14	not acceptable.									
15	8.7. All studs, bolts, nuts shall be coated as per Piping Material Specification (Doc No. P30319-30-99-12-1602).									
16	8.8. Skid shall be provided with shelter to protect the chemical against site temperature.									
17	8.9. Temperature of injection point range from 50 - 85 deg C.									
18	8.10. Vendor shall provide criteria for the sizing calculation of pulsation dampener during detail engineering.									
19	8.11. The Chemical Injection Packages are located in hazardous area Zone 2. However, all instrumentation and valves shall be suitable for									
20	Zone 1 IIB T3, IP66, whereas, electrical equipment including motor shall be suitable for Zone 2 IIB T3.									
21	8.12. Mechanical locking feature for pump stroke adjustment shall be provided.									
22	8.13. Full functional test for the complete package as per specification AGES-SP-05-011 shall be required.									
23	8.14. One set of electric operated chemical drum un-loading pump shall be provided.									
24	8.15. Vendor shall calculate NPSHA value w.r.t. tank and pump layout.									
25	8.16. Electrical requirements shall comply with specification AGES-SP-02-014.									
26	8.17. Refer to Structural Design Basis for Seismic and Wind Design Requirements (Doc No. AGES-SP-01-003).									
27	8.18. Vendor shall update the P&ID and data sheet for Chemical Injection Package for review and approval during detail engineering.									
28	These shall be submitted for Company review and approval as well.									
29	8.19. Chemical injection pump selection shall be based on the following criteria & condition (refer Table-B).									
30	(a) Chemical Injection Package shall be installed with one skid mounted vertical storage tank including required connection between									
31	tank and pump.									
32	(b) Corrosion Inhibitor skid shall be installed with one working pump and one warehouse spare pump shall be supplied.									
33	(c) Skid shall be installed with single-head pumps to cater for the required injection points. Estimated skid size & weight:									
34	(U-045P-01) 2.5 (m) W x 3 (m) L x 2 (m) H - 2 Ton / (U-6230-01) 2.5 (m) W x 3 (m) L x 2 (m) H - 2 Ton.									
35	Vendor shall optimize the skid size, but shall not exceed the indicated values.									
36	(d) Chemical inhibitor dosing concentration is 10ppmv per pump.									
37	8.20 PRVs and PSV inside the skid shall be provided by VENDOR as per the quantity required in P&ID (P30319-44-21-08-1637).									
38	8.21 Package criticality shall be rating 2, inspection class 2 and material certification type 3.1 as per BS EN 10204.									
39	8.22 The quality control requirement shall be in accordance with the project quality document, "Criticality Rating Specification"									
40	Doc no. AGES-SP-13-001 and "Project Procurement and Certification Requirement" Doc no. AGES-SP-13-002.									
41	8.23 Inspection and test shall be complied with the requirements defined in specification AGES-SP-05-006 and AGES-SP-13-002.									
42	8.24 Welding and NDE shall be performed as per Welding & Non-Destructive Examination (NDE) Specification (AGES-SP-07-007).									
43										
44	Table-B									
45										
46	Skid Tag.	Injection Point Location @	No.of. Pump Heads	Pump Flowrate Normal (LPH)	Pump Flowrate Design (LPH)	Pump Discharge Pressure (barg)	Pump Design Pressure (barg)	Tank capacity (working / nominal), m3	Storage Tank dim. (ID x TL/TL (Horz.) m	Remarks
47										
48										
49										
50										
51	65-21-U-6230-01	CLS30(DD) Dual Wellhead Oil Producer	1	0.2	0.28	220	290	0.13 / 0.29	0.5 ID x 1.5 TL/TL (VTC)	Design pressure (barg): 290 Rating: 2500 #
52										
53										
54										
55										
56	65-72-U-045P-01	CLS28(BB) Dual Wellhead Oil Producer	1	0.2	0.28	220	290	0.13 / 0.29	0.5 ID x 1.5 TL/TL (VTC)	Design pressure (barg): 290 Rating: 2500 #
57										
58										
59										
60	* VTC = Vendor to confirm									
61										
62										
63										
64										
65										
66										
67										
68										
69										
70										

MECHANICAL DATA SHEET FOR CORROSION INHIBITOR PACKAGE(CLS-28 BB & CLS-30 DD) AT SHANAYEL SN-45 & SN-40

Project No. P30319

REV.: B

PAGE: 9 OF 10

Document No.: P30319-65-21-34-1601

Date: 05-Jan-2024



LUBRICATION REQUIREMENTS

1	APPLICABLE TO:	☒ PROPOSAL	☐ PURCHASE	☐ AS BUILT	PACKAGE TAG NO.	65-72-U-045P-01 / 65-21-U-6230-01	REVISION
2	CLIENT	ADNOC ONSHORE			SERVICE	Corrosion Inhibitor Injection	
3	LOCATION / FACILITY	CLS 28(BB) / CLS 30(DD) SHANAYEL					
4	PACKAGE MANUFACTURER	As per approved Vendor list			MODEL	By Vendor	
5					SERIAL NO.	By Vendor	
6	EQUIPMENT TYPES	Pump			NO. OF UNITS	2 per package	
7	PARTS REQUIRING LUBRICATION	Bearings					
8	LUBRICANT TYPE (OIL/GREASE)	Oil					
9	LUBE SPECIFICATION						
10	RECOMMENDED VISCOSITY (SUMMER)		cP	°C (WINTER)		cP	°C
11	INITIAL FILL CAPACITY			EST. ANNUAL FILL CAPACITY			
12	PERIODIC REQUIREMENT			FREQUENCY			
13							
14	LUBRICANT TRADE NAMES						
15							
16	EQUIPMENT TYPES	Motor			NO. OF UNITS	2 per package	
17	PARTS REQUIRING LUBRICATION	Bearings					
18	LUBRICANT TYPE (OIL/GREASE)						
19	LUBE SPECIFICATION						
20	RECOMMENDED VISCOSITY (SUMMER)		cP	°C (WINTER)		cP	°C
21	INITIAL FILL CAPACITY			EST. ANNUAL FILL CAPACITY			
22	PERIODIC REQUIREMENT			FREQUENCY			
23							
24	LUBRICANT TRADE NAMES						
25							
26	EQUIPMENT TYPES				NO. OF UNITS		
27	PARTS REQUIRING LUBRICATION						
28	LUBRICANT TYPE (OIL/GREASE)						
29	LUBE SPECIFICATION						
30	RECOMMENDED VISCOSITY (SUMMER)		cP	°C (WINTER)		cP	°C
31	INITIAL FILL CAPACITY			EST. ANNUAL FILL CAPACITY			
32	PERIODIC REQUIREMENT			FREQUENCY			
33							
34	LUBRICANT TRADE NAMES						
35							
36	EQUIPMENT TYPES				NO. OF UNITS		
37	PARTS REQUIRING LUBRICATION						
38	LUBRICANT TYPE (OIL/GREASE)						
39	LUBE SPECIFICATION						
40	RECOMMENDED VISCOSITY (SUMMER)		cP	°C (WINTER)		cP	°C
41	INITIAL FILL CAPACITY			EST. ANNUAL FILL CAPACITY			
42	PERIODIC REQUIREMENT			FREQUENCY			
43							
44	LUBRICANT TRADE NAMES						
45							
46	NOTES:						
47							
48							
49							
50							
51							
52							
53							
54							
55							
56							
57							
58							
59							
60							
61							
62							
63							
64							
65							
66							
67							

MECHANICAL DATA SHEET FOR CORROSION INHIBITOR PACKAGE(CLS-28 BB & CLS-30 DD) AT SHANAYEL SN-45 & SN-40

Project No. P30319 REV.: B PAGE: 10 OF 10
Document No.: P30319-65-21-34-1601 Date: 05-Jan-2024



NOISE DATA SHEET

1	APPLICABLE TO: ☒ PROPOSAL ☐ PURCHASE ☐ AS BUILT		PACKAGE TAG NO. 65-72-U-045P-01 / 65-21-U-6230-01		REVISION						
2	CLIENT ADNOC ONSHORE		SERVICE Corrosion Inhibitor Injection								
3	LOCATION / FACILITY CLS 28(BB) / CLS 30(DD) SHANAYEL										
4	PACKAGE MANUFACTURER As per approved Vendor list		MODEL	By Vendor SERIAL NO. By Vendor							
5											
6	EQUIPMENT	DESCRIPTION	NUMBER	ITEM	PROVIDED	MOUNTED					
7	CENTRIFUGAL										
8	RECIPROCATING										
9	ROTARY										
10	GEAR										
11	MOTOR	Injection Pump Motor		as per data sheet	Vendor	Vendor					
12	PUMP	Injection Pump		as per data sheet	Vendor	Vendor					
13	GAS TURBINE										
14	STEAM TURBINE										
15	FAN										
16	OTHER										
17	MAXIMUM (SPL) SOUND PRESSURE LEVELS (dBA) @ 1m										
18	EQUIPMENT ITEM	PURCHASER'S REQUIREMENT	VENDOR'S PREDICTE	VENDOR'S PREDICTED @ OCTAVE BAND CENTER							
19				62.5	125	250	500	1000	2000	4000	8000
20	COMPLETE	<85									
21	DRIVEN										
22	DRIVER										
23	GEAR										
24	OTHER										
25	NOTES:										
26											
27											
28											
29											
30											
31											
32											
33											
34											
35											
36											
37											
38											
39											
40											
41											
42											
43											
44											
45											
46											
47											
48											
49											
50											
51											
52											
53											
54											
55											
56											
57											
58											
59											
60											
61											
62											
63											
64											
65											
66											
67											
68											